

Solution to Tutorial exercises: Week 3.

Exercise: PV Hosting Capacity using the Test LV Circuit

Note: The provided code fixes the random seeds for the load profiles and PV allocations; however, it does not fix the randomness associated with power factor, so your results may differ slightly. The key point is that the power-factor variations are small, so you may see a mismatch of approximately ± 2 customers compared with the results shown in this solution.

E.1: Starting from 0% PV penetration, increase the penetration by 20%, up to 100% PV penetration for each of the specified days per season (see below) using a worst-case scenario PV profile of the corresponding season. For each PV penetration and season, quantify how many customers have voltages that are non-compliant.

- **Summer Day: start_day=21**
- **Autumn Day: start_day=59**

Solution:

First, you need to identify the curve corresponding to the worst-case scenario. That is, the PV profile that is expected to produce the most severe voltage issues.

After running the simulations, the following results are obtained:

Season	Penetration level					
	0%	20%	40%	60%	80%	100%
Summer	0	0	4	16	25	27
Autumn	0	0	0	12	17	25

To illustrate the workflow in more detail, Figures 1 and 2 show the transition from 20% to 40% PV penetration in the Summer case. At 20% penetration there are no voltage issues, whereas at 40% some customers exceed the 253 V limit. To quantify non-compliance, record the customer voltages throughout the daily simulation and then use the code to count how many customers exceed the limit at least once.

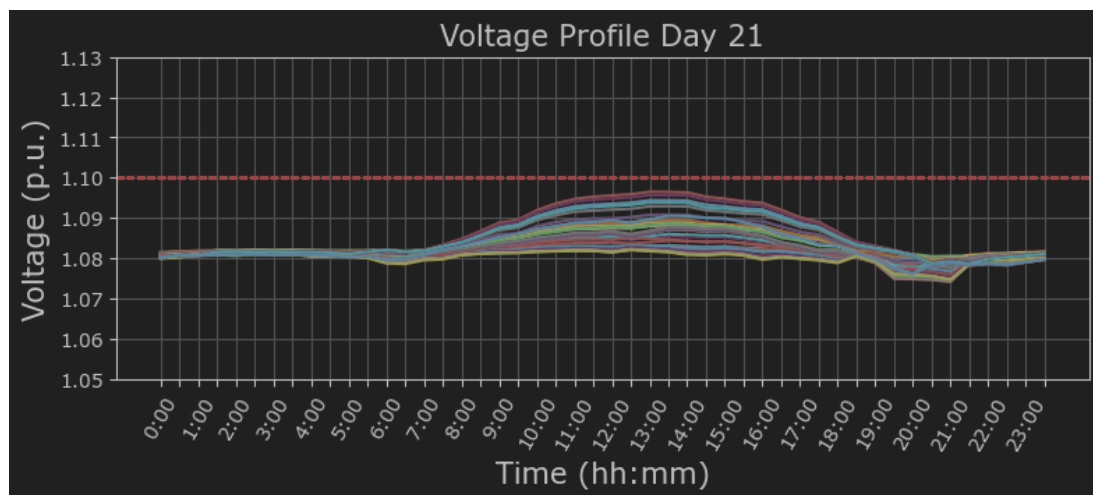


Figure 1. Summer case (20% PV penetration): customer voltage profile—no overvoltage events.

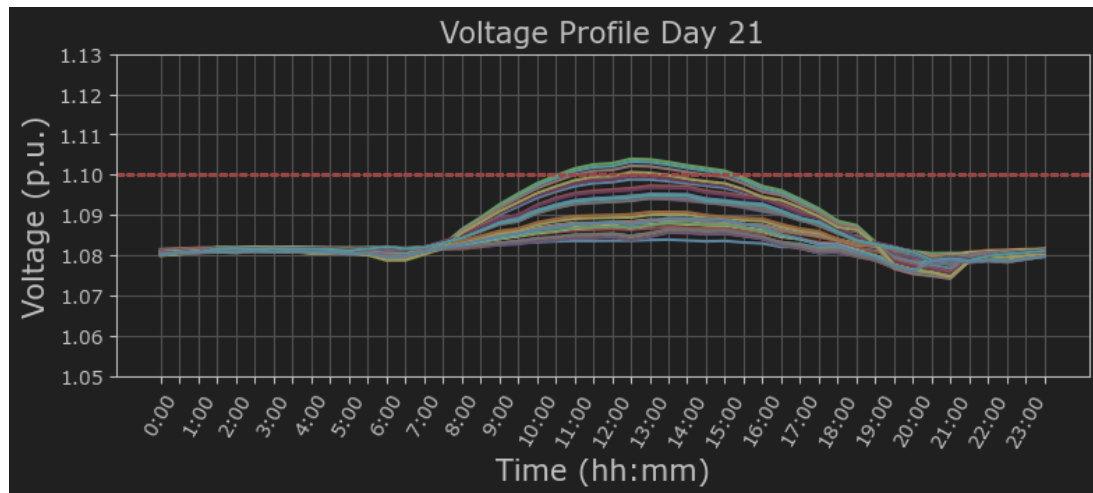


Figure 2. Summer case (40% PV penetration): customer voltage profile—example overvoltage events above 253 V.

E.2: For each PV penetration and season from E.1, extract the active (P) power and reactive (Q) power for all the three phases of the transformer. Compute the apparent power ($|S|$) in kVA and check the level of utilisation of the transformer assuming a rated capacity of 175kVA.

Solution:

After running the simulations, the transformer utilisation is:

Season	Penetration level					
	0%	20%	40%	60%	80%	100%
Summer	15.8%	24.9%	54.0%	88.4%	118%	147 %
Autumn	13.5%	18.0%	42.7%	72.2 %	97.9%	123 %

The transformer utilisation results above introduce the key issue of *reverse power flow*. As PV penetration increases, local generation can exceed local demand (particularly around midday), so power is exported upstream toward the transformer, raising the apparent power it must carry rather than reducing it.

This effect is already visible in Autumn when moving from 0% to 20% PV penetration: utilisation increases from 13.5% (23.63 kVA) to 18.0% (31.5 kVA). Recall that each PV system is rated at 10 kVA; at 20% penetration there are 6 customers with PV, giving an aggregate midday injection of approximately 55–60 kW. Taking the magnitude of the net flow as $|23.63 \text{ kVA} - 55 \text{ kVA}| = 31.37 \text{ kVA}$ (approx.), we obtain a value close to the 31.5 kVA (18% utilisation), as reported in the table.

This contrasts with the previous tutorial (a different network), where the main observable effect of increasing PV penetration was a reduction in losses, suggesting a lower utilisation level. Here, the case is more extreme: PV exports are large enough that reverse power flow becomes the dominant effect, increasing the apparent power through the transformer and raising its utilisation. As penetration increases further, this export-driven loading can significantly increase feeder losses and may even exceed the transformer's rated capacity.